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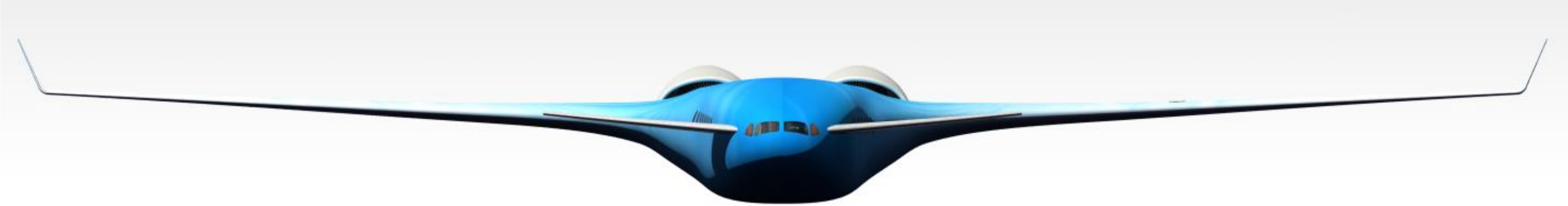

AE4238: Aero Engine Technology

Engine Inlets-III: Installation Effects

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TU Delft

10/21/2022

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What will we learn in this lecture

- What are the effects of engine installation
- Flowfield around a nacelle
- Design considerations

Engine development

- Boeing 737-100 (1967)
  - 2x Pratt & Whitney JT8D turbofan
  - Bypass ratio 1
  - SFC 22.7 g/kN.s

- Boeing 737-400 classic (1988)
  - CFM56-3 series
  - Bypass ratio 5.1
  - SFC 18.9 g/kN.s

Engine development

- Boeing 737-800 Next generation (1997)
  - CFM56-7 series
  - Bypass ratio 5.1-5.3
  - SFC 17.8 g/kN.s

- Boeing 737 MAX (2017)
  - Leap 1B
  - Bypass ratio 9
  - SFC 14.9 g/kN.s

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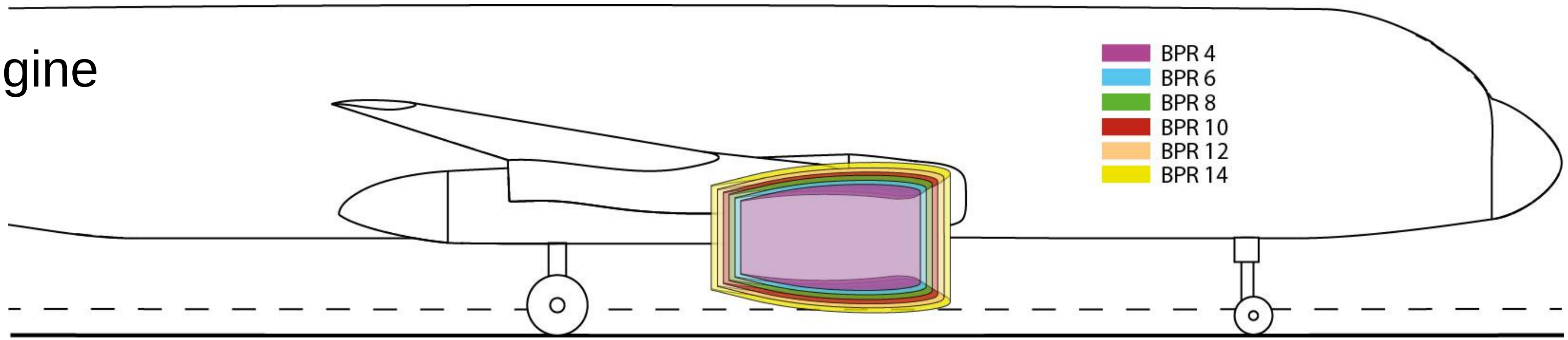
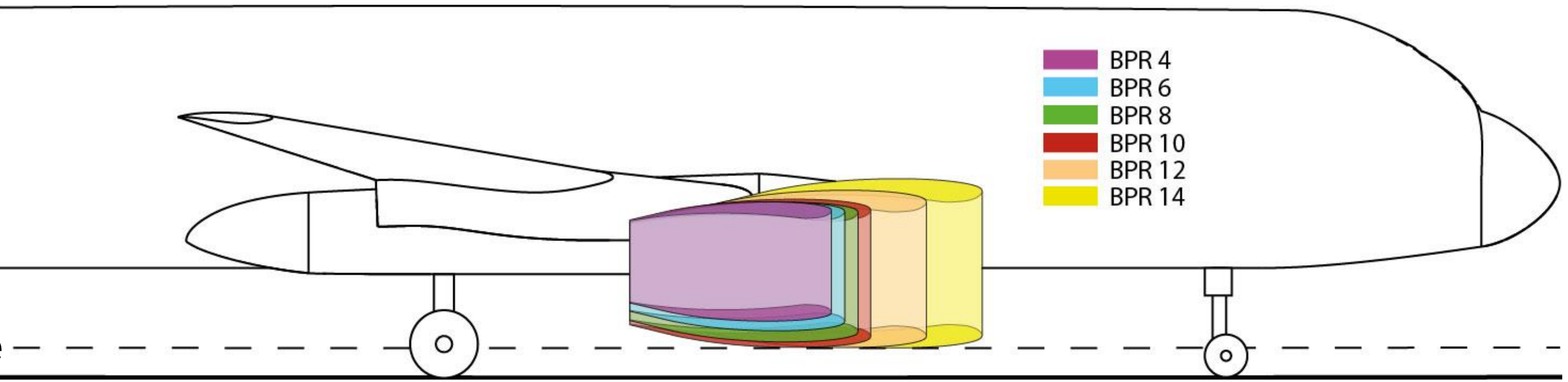
Engine locations

Case 1:
Constant engine location



Case 2:
Fixed spanwise location

- Preserve ground clearance
- Preserve gully distance



Flow around a Nacelle

Picture: DNW

Drag of a Nacelle

1. Profile Drag
2. Spillage Drag
3. Wave Drag
4. Windmilling Drag

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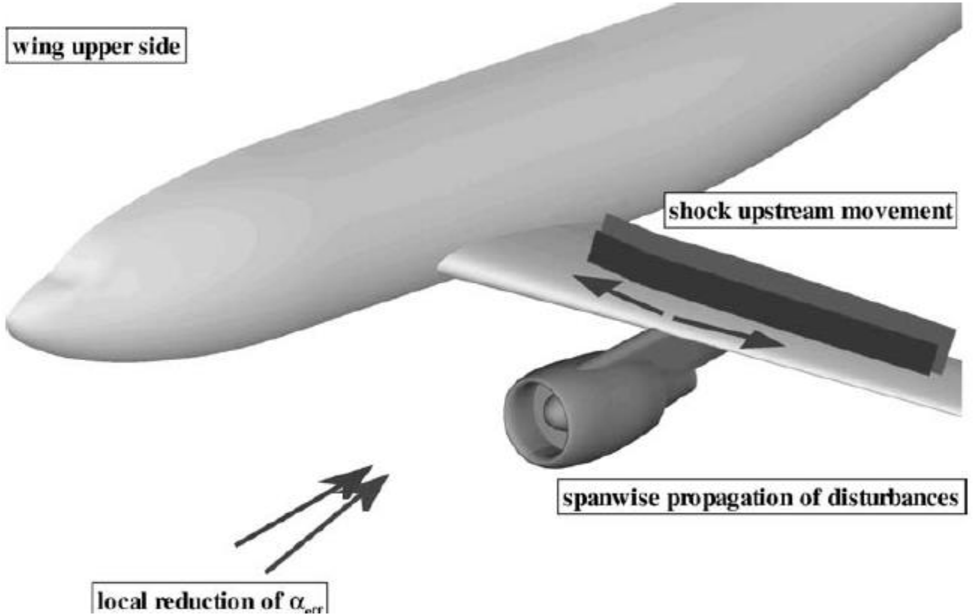
Flow around a nacelle



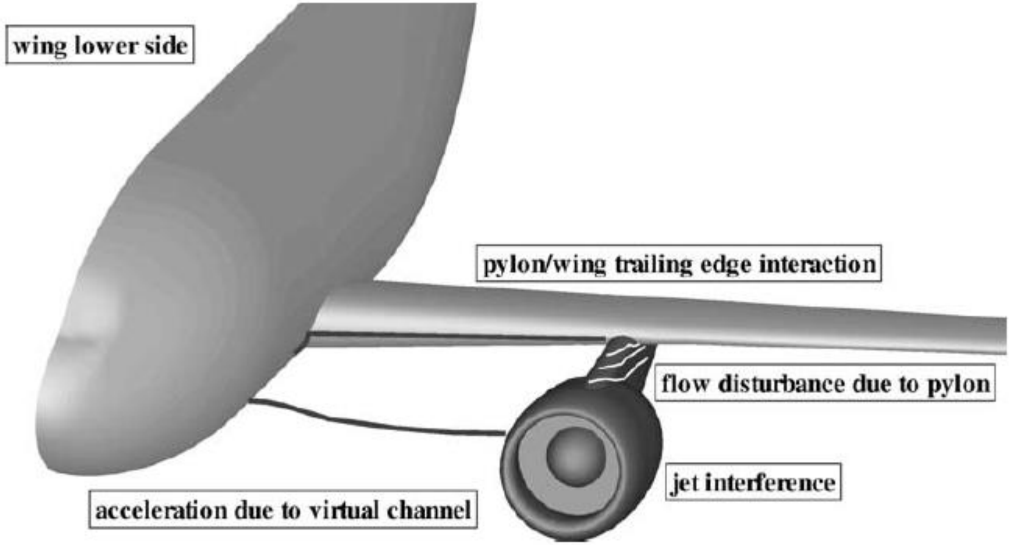


Fig. 3. Flow phenomena due to engine/airframe interference.

Ralf Rudnik, Cord-Christian Rossow, Heiko Frhr. v. Geyr
“Numerical simulation of engine/airframe integration for high-bypass engines”, Aerospace Science and Technology 6 (2002) 31–42

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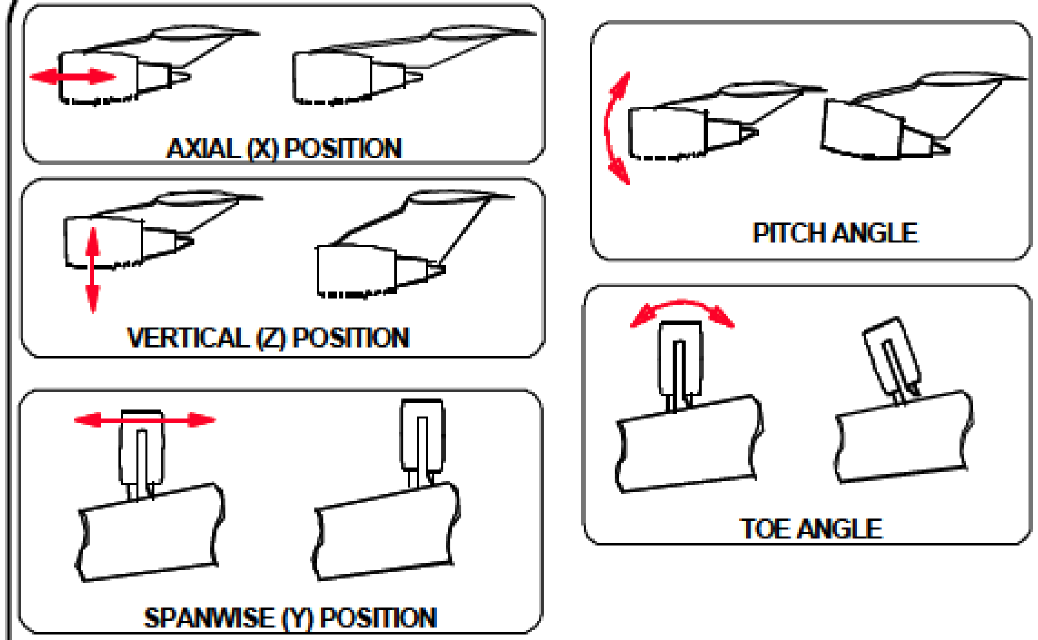
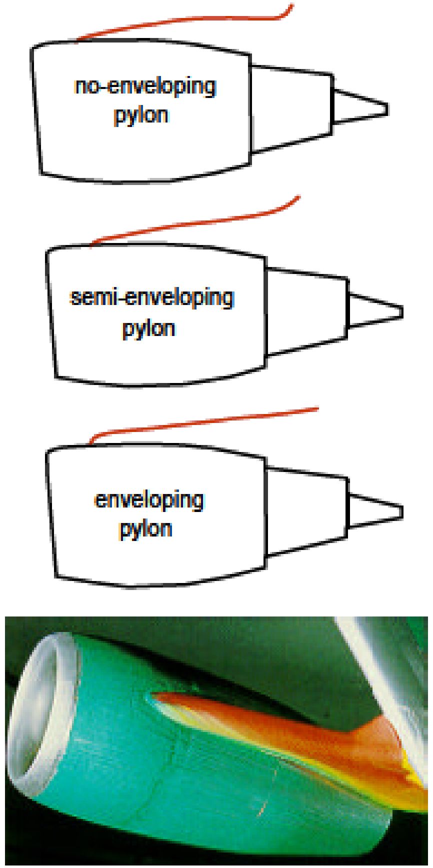
Engine Integration



Engine position on wing



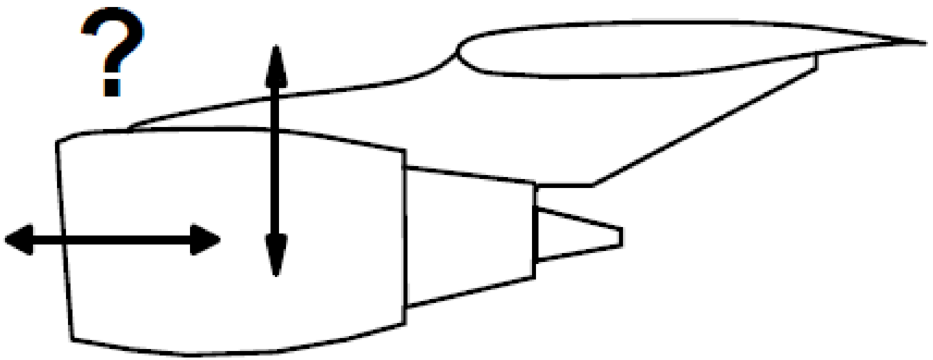
Pylon and nacelle external shapes

Where to place the engine?

To satisfy following criteria

Engine should

Ground clearance	Move upward
Static and dynamic loads	Move backward
Turbine burst	Move forward
Wing bending relief	Move outboard
Lateral control (engine failure)	Move inboard
Noise reduction	Long cowl design
Drag minimization	Move downward and forward ; Short cowl design



Slide from Airb

Various drag components

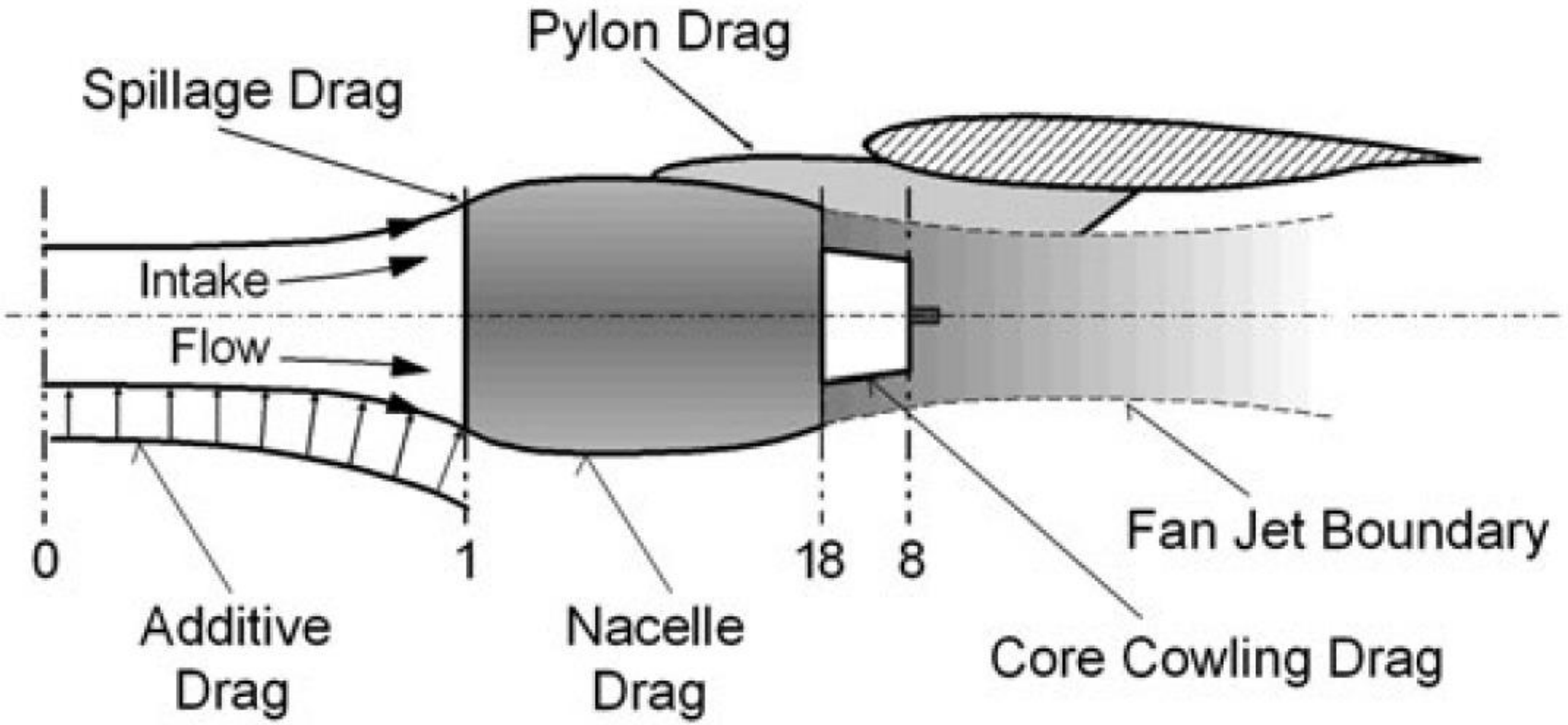


Fig. 15 Installed engine drag components

H. Hoheisel, H. Frhr. von Geyr "The influence of engine thrust behaviour on the aerodynamics of engine airframe integration" CEAS Aeronaut J (2012) 3:79–92

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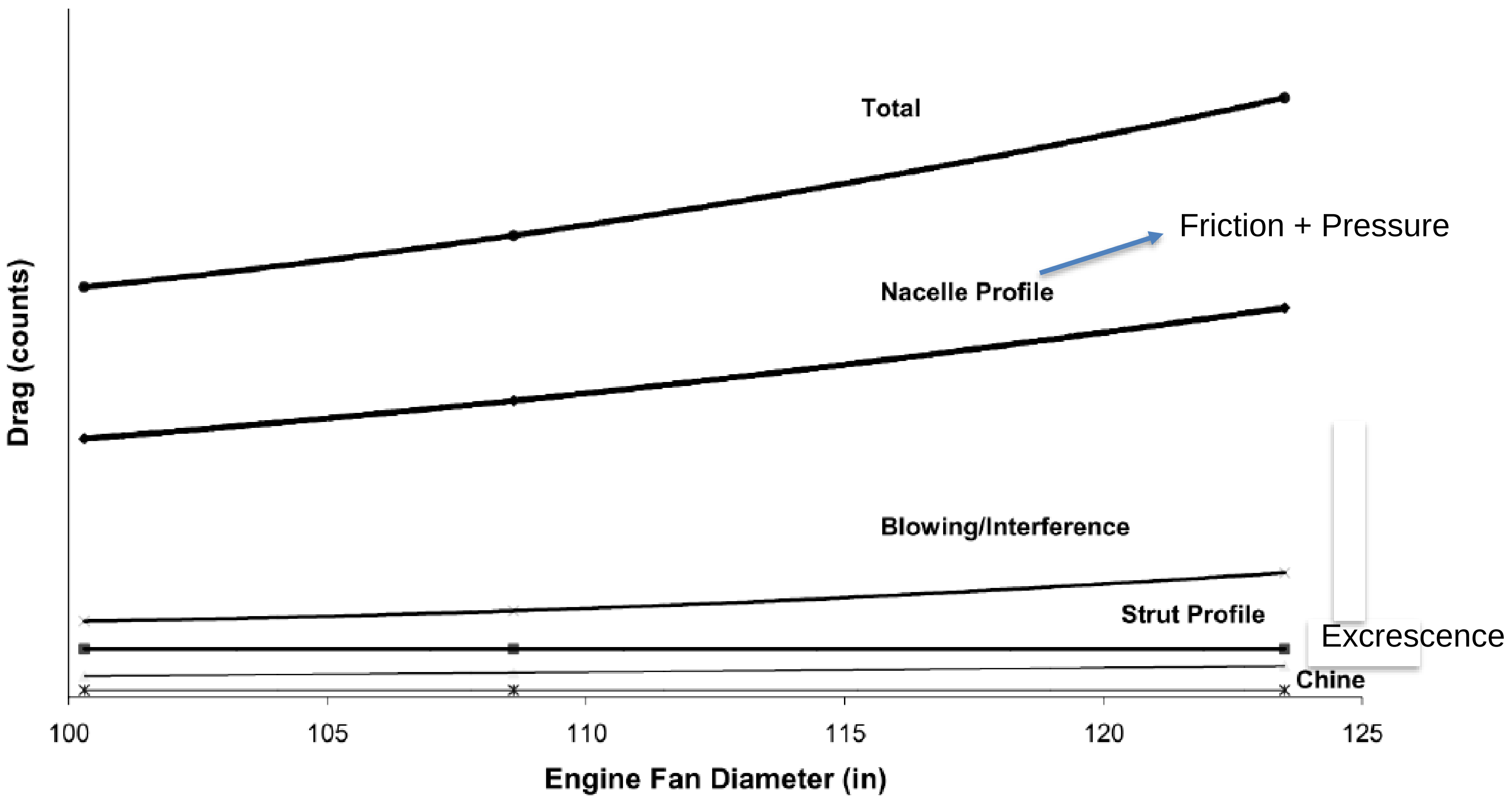
Nacelle drag breakdown



David L Daggett, Stephen T Brown, and Ron T Kawai. Ultra-efficient engine diameter study. Technical report, NASA, 2003.

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Interference

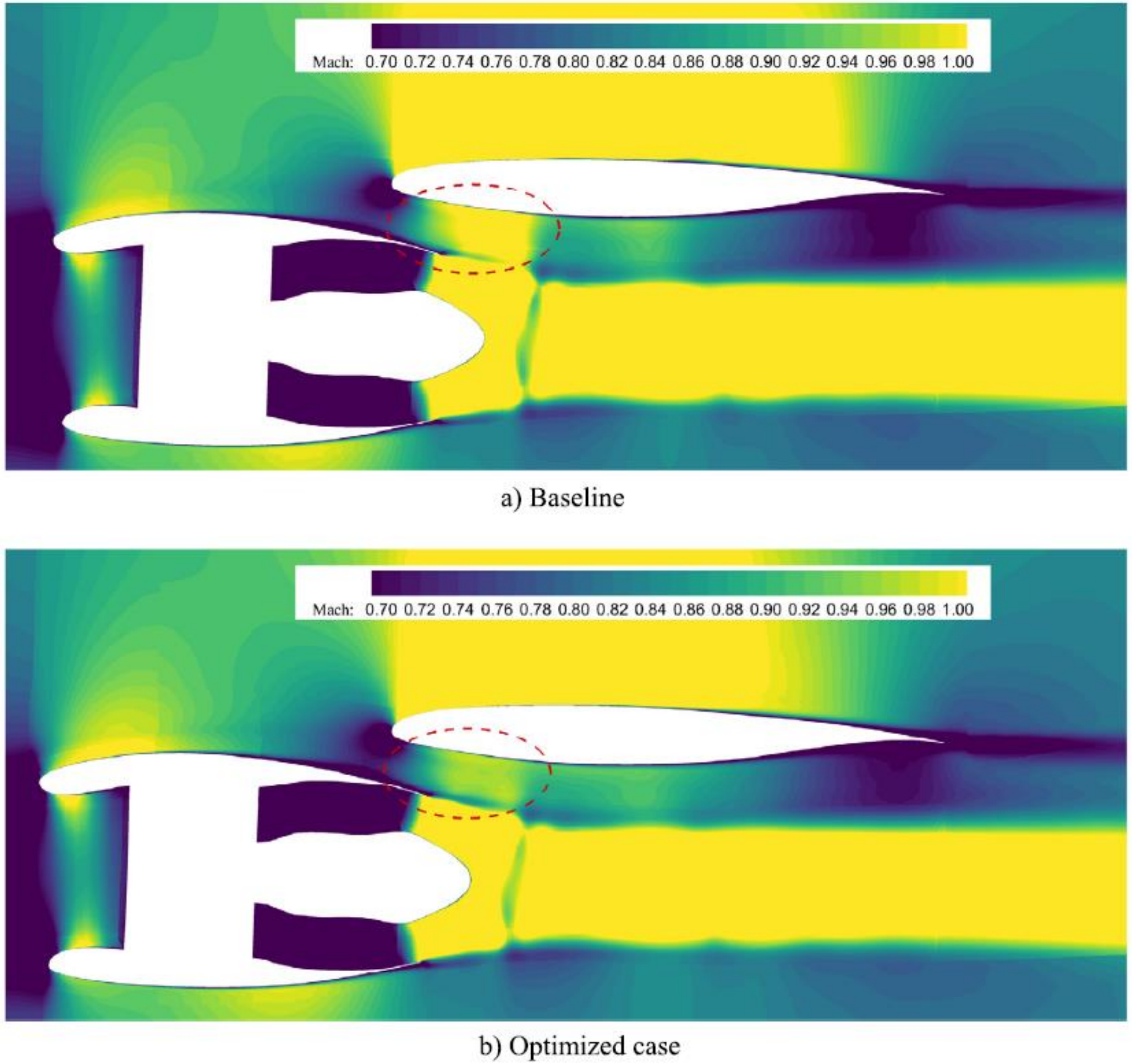

Fig. 14. Counter of Mach number at 29.5% half span.

RuiwuLei, JunqiangBai,∗, DanyangXu "Aerodynamic optimization of civil aircraft with wing-mounted engine jet based on adjoint method", Aerospace Science and Technology 93 (2019) 105285

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Wind tunnel experiments

Fig. 1 The ALVAST model equipped with different engine simulators in the test section of the ONERA S1MA wind tunnel. *Left* turbofan, *middle* VHBR, *right* UHBR simulator

H. Hoheisel, H. Frhr. von Geyr "The influence of engine thrust behaviour on the aerodynamics of engine airframe integration" CEAS Aeronaut J (2012) 3:79–92

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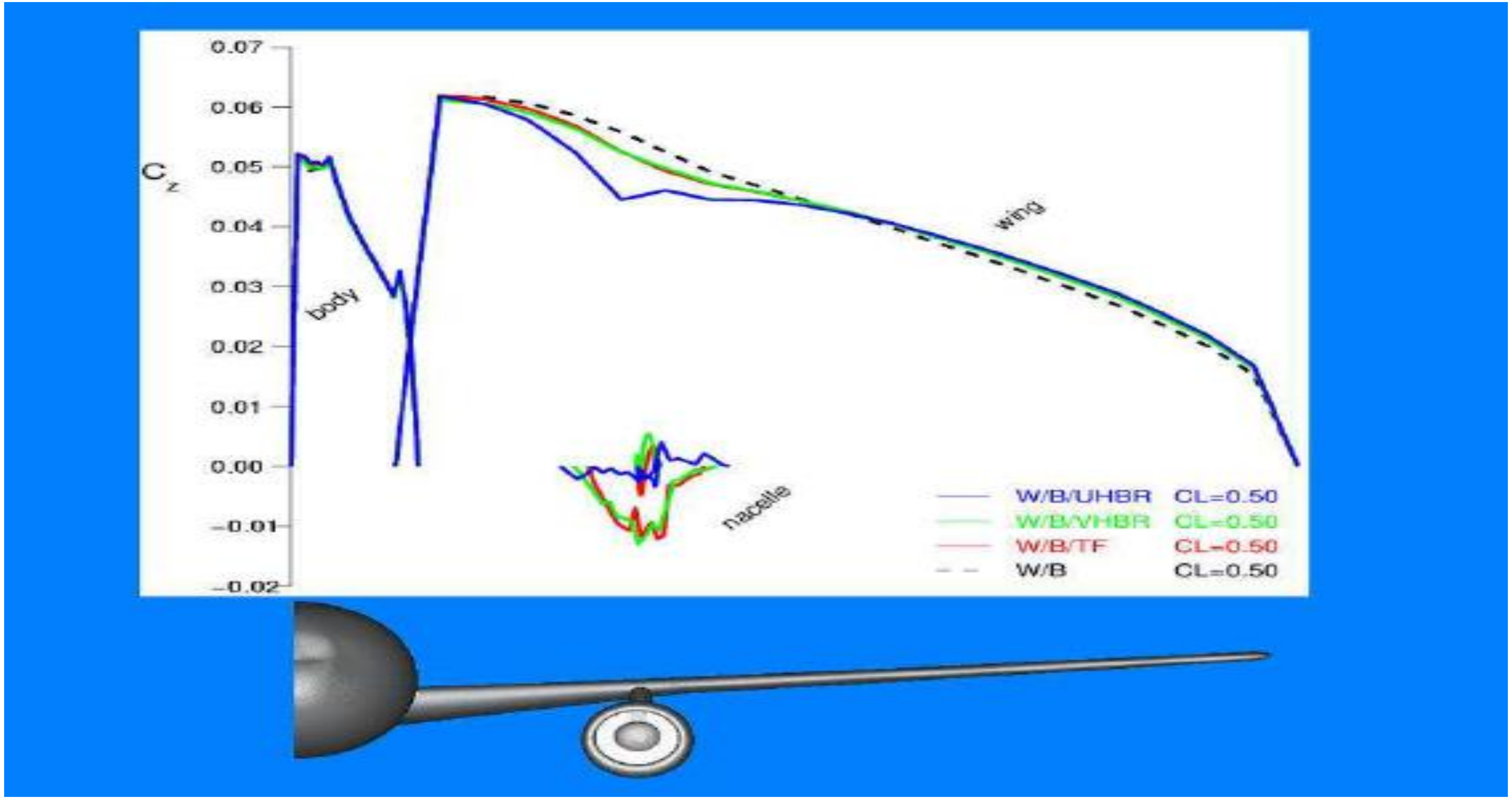
Effect of engine installation on spanwise loading



Slide: Prof. G. Eitel

Effect of engine installation on drag

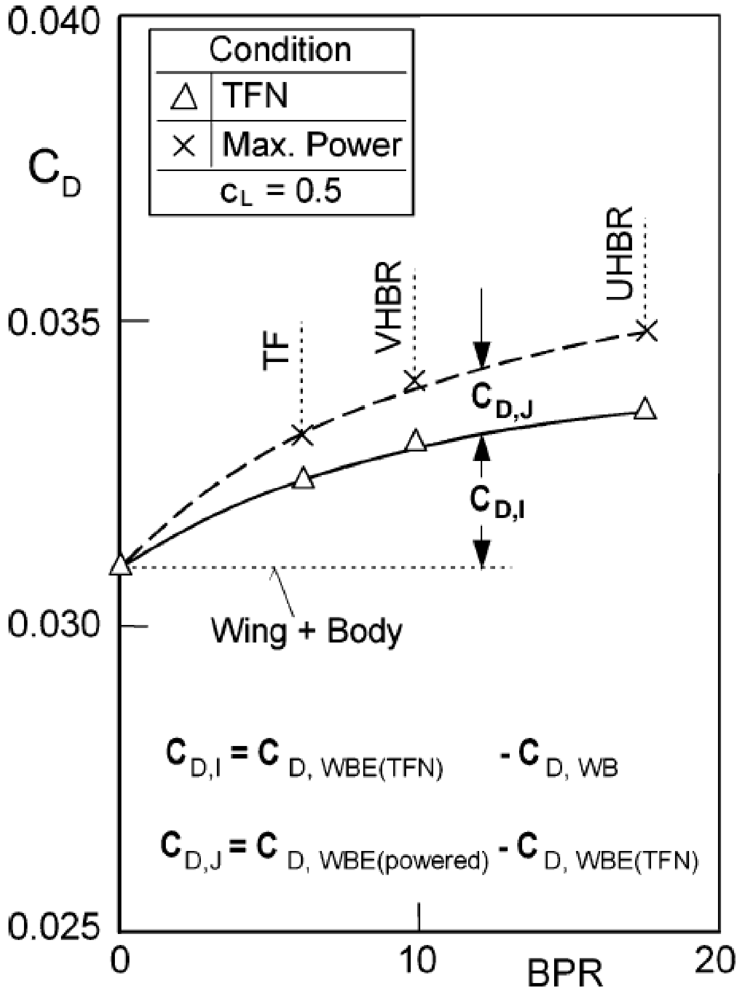
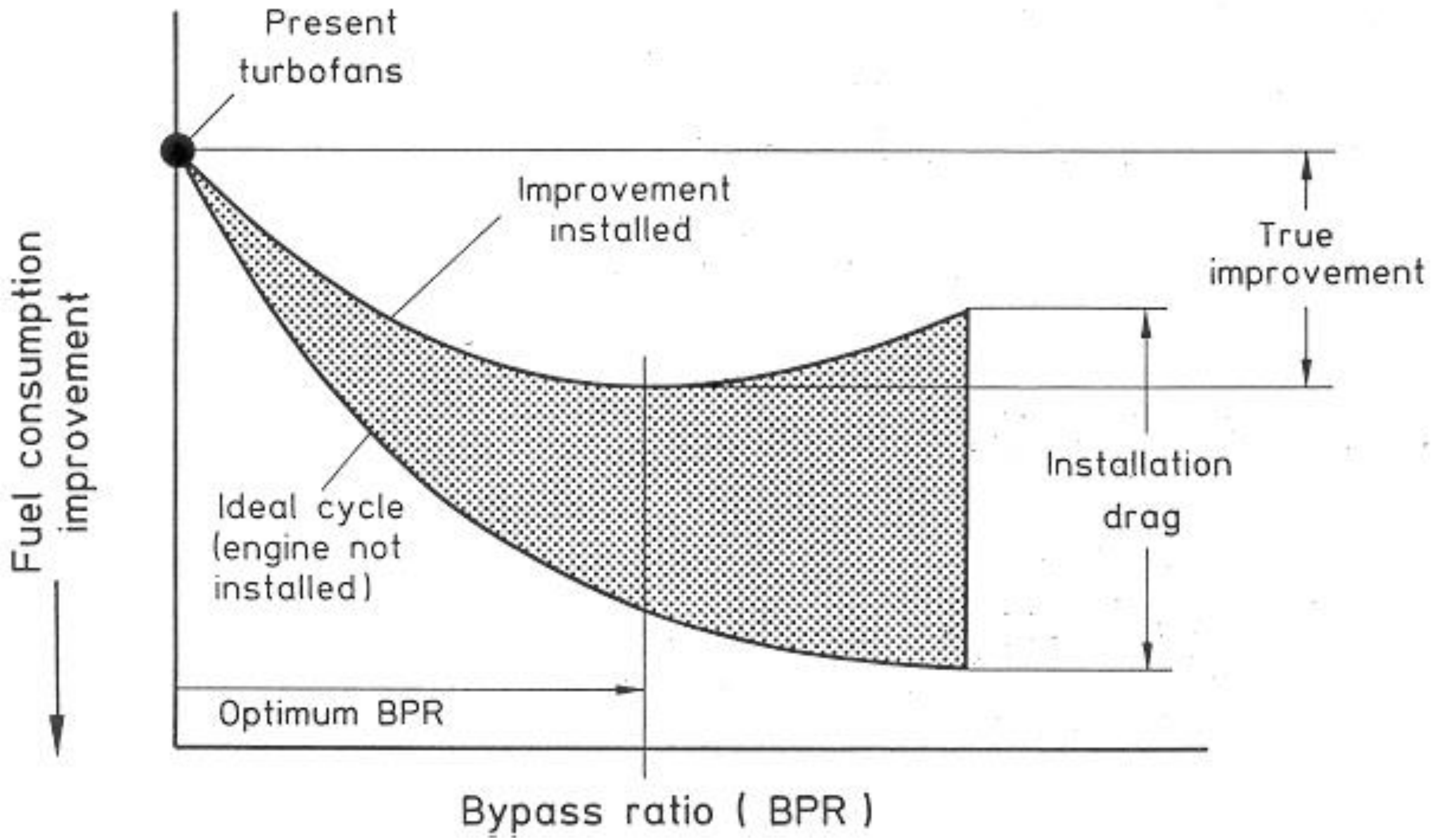


Fig. 5 Influence of the engine bypass ratio on the measured drag coefficient of the wing/body/pylon/engine configuration at $Ma_0 = 0.75$

H. Hoheisel, H. Frhr. von Geyr “The influence of engine thrust behaviour on the aerodynamics of engine airframe integration” CEAS Aeronaut J (2012) 3:79–92

Effect of engine installation



Ralf Rudnik, Cord-Christian Rossow, Heiko Frhr. v. Geyr "Numerical simulation of engine/airframe integration for high-bypass engines", Aerospace Science and Technology 6 (2002) 31–42

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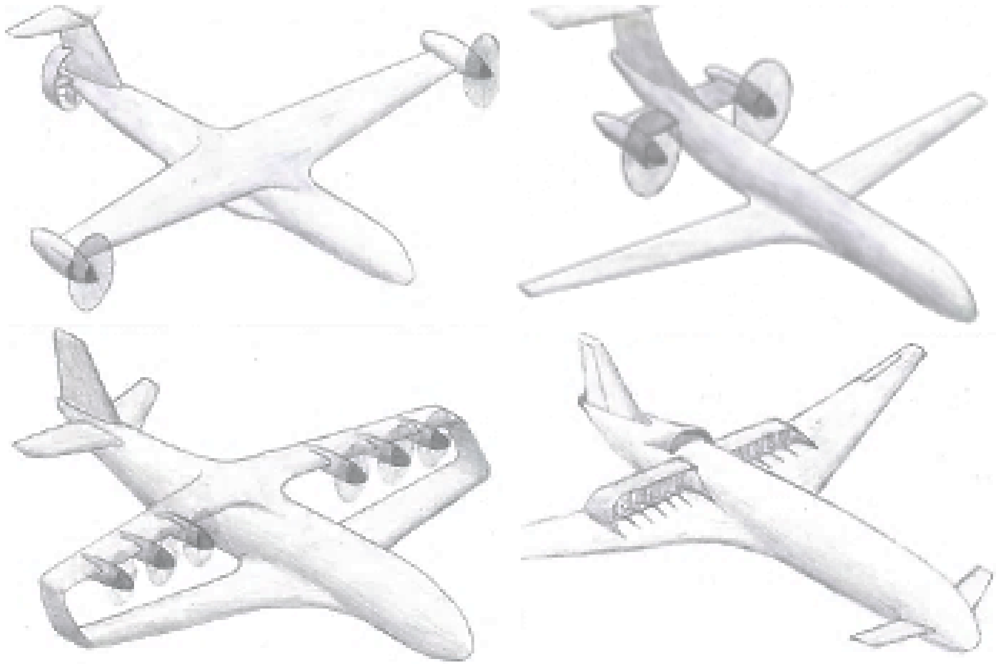
Thank You



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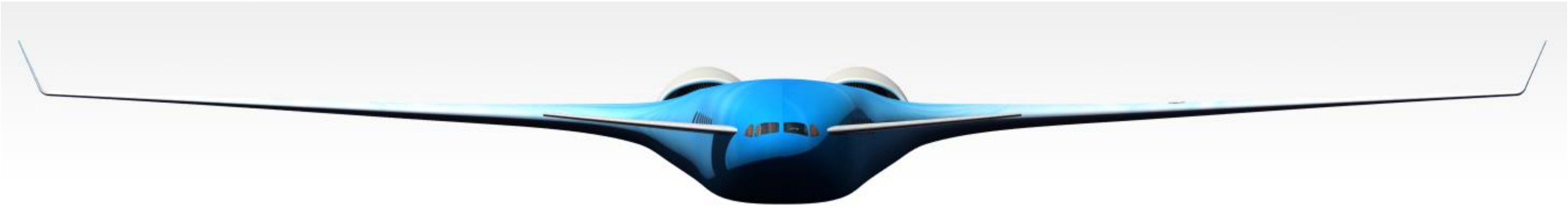

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